

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287717

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Kao; Min-Feng et al.

IMAGE SENSOR DEVICE AND SEMICONDUCTOR STRUCTURE

Abstract

An image sensor device includes a first semiconductor structure and a second semiconductor structure bonding with the first semiconductor structure. The first semiconductor structure includes a first semiconductor substrate and logic devices. The logic devices are disposed on the first semiconductor substrate. A through-substrate via is penetrating through the first semiconductor substrate and located between two of the logic devices. A second semiconductor structure includes a second semiconductor substrate and image sensors. The image sensors are disposed in a pixel area of the second semiconductor substrate. At least one of the image sensors is electrically to at least one of the two of the logic devices.

Inventors: Kao; Min-Feng (Chiayi City, TW), Lin; Hsing-Chih (Tainan City, TW), Liu; Jen-Cheng (Hsin-Chu City, TW), Yaung; Dun-Nian (Taipei City, TW)

Applicant: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

Family ID: 1000007726444

Assignee: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

Appl. No.: 18/596667

Filed: March 06, 2024

Publication Classification

Int. Cl.: H01L27/146 (20060101)

U.S. Cl.:

CPC H10F39/811 (20250101); H10F39/182 (20250101);

Background/Summary

BACKGROUND

[0001] The semiconductor integrated circuit (IC) industry has witnessed an exponential surge in growth. This growth has been fueled by technological advancements in IC materials and design, leading to the production of successive generations of ICs. Each generation is characterized by smaller, more intricate circuits than its predecessor. As ICs have evolved, there has been a general increase in functional density (i.e., the number of interconnected devices per chip area), while the geometry size (i.e., the smallest component or line that can be fabricated) has decreased. This miniaturization process typically yields benefits such as enhanced production efficiency and reduced costs. However, it also escalates the complexity of IC processing and manufacturing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. 1A is a schematic cross-sectional view of an embodiment image sensor device in accordance with various embodiments.

[0004] FIG. 1B is a schematic top view of a first semiconductor structure in accordance with various embodiments.

[0005] FIG. 1C is a schematic top view of a second semiconductor structure in accordance with various embodiments.

[0006] FIGS. 2A to 2J are schematic cross-sectional views of a fabrication method of an embodiment image sensor device in accordance with various embodiments.

[0007] FIG. 3 is a schematic cross-sectional view of an embodiment first semiconductor structure in accordance with various embodiments.

[0008] FIGS. 4A to 4H are schematic cross-sectional views of a fabrication method of an embodiment image sensor device in accordance with various embodiments.

[0009] FIGS. 5A to 5N are schematic cross-sectional views of a fabrication method of an embodiment image sensor device in accordance with various embodiments.

[0010] FIG. 6 is a schematic cross-sectional view of an embodiment first semiconductor structure in accordance with various embodiments.

DETAILED DESCRIPTION

[0011] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0012] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are

intended to encompass different orientations of the structure in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0013] In some integrated circuits, the front side of a substrate of a wafer includes various doped regions. These doped regions may work with the gate stack structures formed over the substrate to function as transistors. An interlayer dielectric (ILD) layer may be disposed above the front side of the substrate and cover the gate stack structures. The ILD layer contains various contact vias that connect to the transistors, such as gate contact vias and source/drain contact vias. An interconnect structure may be disposed above the ILD layer and include multiple metallization layers and inter-metal dielectric (IMD) layers situated between the metallization layers. The metallization layers in the interconnect structure may be referred to as M1, M2, M3 . . . Mn layers, depending on their level. The metallization layer closest to the ILD layer may also be referred to as the M1 layer, while the metallization layer furthest from the ILD layer may also be referred to as the Mn layer.

[0014] In some integrated circuits, a backside circuit structure is formed on the backside of the substrate, and a through-substrate via (TSV), also known as a backside through-substrate via (BTSV) in some cases, is used to connect the backside circuit structure with the interconnect structure over the front side of the substrate. In certain integrated circuits, the TSV extends from the backside of the substrate to the interconnect structure, potentially reaching as far as the M1 layer. This necessitates the TSV to traverse entirely through the ILD layer that lies between the interconnect structure and the substrate. As a consequence, the TSV is required to possess a significantly larger depth and area.

[0015] In some embodiments disclosed here, the depth and size of the TSV are reduced, thereby providing greater design flexibility of the integrated circuits. For instance, in an image sensor device, to avoid any negative impact on resolution, the TSV is typically positioned around the periphery of the pixel area of the image sensor device, ensuring that it does not overlap with the radiation sensing elements in the pixel area. However, if the size of the TSV can be reduced, the TSV may overlap with the pixel area without affecting the resolution of the image sensor device. This not only shortens the signal path from the radiation sensing elements to the TSV but also reduces the size of the image sensor device. By shortening the signal path between the radiation sensing element and the TSV, not only can the resistance-capacitance delay (RC delay) of the signal be reduced, but it also minimizes the issue of signal crosstalk during transmission.

[0016] FIG. 1A is a schematic cross-sectional view of an embodiment image sensor device **1** in accordance with various embodiments. The image sensor device **1** includes a first semiconductor structure **10** and a second semiconductor structure **20**. The first semiconductor structure **10** and the second semiconductor structure **20** can either be dies or wafers. Alternatively, one of them could be a die while the other is a wafer. In some embodiments, both the first semiconductor structure **10** and the second semiconductor structure **20** can be System-on-Chip (SoC) dies or other appropriate semiconductor dies.

[0017] Referring to FIGS. 1A and 1B, the first semiconductor structure **10**, alternatively referred to as a logic wafer (or logic die), includes a variety of components such as read-out circuits, decoders, registers, multiplexers/de-multiplexers, amplifiers, reference pixels, and so on. In some instances, the first semiconductor structure **10** may include an application-specific integrated circuit (ASIC).

[0018] The first semiconductor structure **10** comprises multiple transistors, denoted as T1. These transistors are located in an active region **11** of the first semiconductor structure **10** and are arranged in multiple rows and columns. For instance, the first semiconductor structure **10** includes read-out circuits ROC, and each read-out circuit ROC incorporates several logic devices including the transistor T1. In FIG. 1B, the transistors T1 in the first semiconductor structure **10** are shown, while other types of transistors are not depicted. For example, the transistors T1 may be logic devices, such as reset transistors, source follower transistors, or row selector transistors.

[0019] In some implementations, the read-out circuits within the same row are electrically connected to each other, while the read-out circuits in the same column are electrically connected to multiple through-substrate vias (TSVs) **12**. For example, the transistors **T1** in one column correspond to multiple TSVs **12**. These TSVs **12** may be located either within or outside the active region **11**. At least one of the TSVs **12** disposed within the active region **11** is located between two adjacent transistors **T1**. In contrast to devices where the TSVs **12** are positioned outside the active region **11**, situating the TSVs **12** between the transistors **T1** can effectively shorten the signal transmission path between the transistors **T1** and their corresponding TSVs **12**. This arrangement not only reduces the Resistance-Capacitance (RC) delay of the signal but also improves the frame rate of the image sensor device **1**. In some embodiments, the distance **D** between the TSV **12** and the farthest one of the corresponding transistors **T1** that is electrically connected to the TSV **12** ranges from 1000 micrometers to 9000 micrometers. In some embodiments, signal lines between the transistors **T1** and the corresponding TSVs **12** can be referred to as bit lines.

[0020] Referring to FIGS. **1A** and **1C**, the second semiconductor structure **20**, alternatively referred to as a sensor wafer (or sensor die), includes image sensors **IS**. The image sensors **IS** are disposed in a pixel area **21** and are arranged in multiple rows and columns. In some embodiments, the image sensors **IS** includes a radiation-sensing region, a transfer gate, and a charge-to-voltage conversion region. The radiation-sensing region is configured to receive and collect a radiation wave that enter from the back side of the second semiconductor structure **20**. Specifically, the radiation-sensing region receives photons of the radiation wave and converts the photons into an electrical signal and charges. The radiation-sensing region is realized as a photodiode, a pinned photodiode, or a p-n junction disposed within the semiconductor substrate of the second semiconductor structure **20**, which is able to generate a signal related to the intensity or brightness of light that strikes the radiation-sensing region. The radiation wave stimulates the radiation-sensing region, leading to the generation of electron-hole pairs in a depletion region of the photodiode. Furthermore, the radiation-sensing region has the capability to sense or detect radiation waves at specific wavelengths, which may be associated with lights of various colors.

[0021] The transfer gate is interposed between the radiation-sensing region and the charge-to-voltage conversion region. The transfer gate is implemented as a control gate having a metal gate structure or a polysilicon structure. The electrons or charges in the radiation-sensing region transfer to the charge-to-voltage conversion region under the control of the transfer gate. Further, the charge-to-voltage conversion region serves as a region configured for transforming the electrons or charges into a corresponding voltage.

[0022] The second semiconductor structure **20** is bonded to the first semiconductor structure **10**. The bonding process can be accomplished through various methods, including but not limited to, die-to-die assemblies, wafer-to-wafer assemblies, die-to-substrate assemblies, and die-to-wafer assemblies. As a result, a three-dimensional (3D) structure is formed. The bonding process can be accomplished through methods such as fusion bonding, hybrid bonding, or other suitable techniques. In some embodiments, the read-out circuits in the first semiconductor structure **10** are electrically connected with the radiation sensing elements **SE** of the second semiconductor structure **20**. The pixel area **21** of the second semiconductor structure **20** is at least partially overlapping with the active region **11** of the first semiconductor structure **10**. In some embodiments, at least a portion of the TSVs **12** and the transistors in the first semiconductor structure **10** are overlapping with the pixel area **21** of the second semiconductor structure **20**.

[0023] FIGS. **2A** to **2J** are schematic cross-sectional views of a fabrication method of an embodiment image sensor device **2** in accordance with various embodiments. Referring to FIG. **2A**, a first semiconductor substrate **100** is provided. The first semiconductor substrate **100** may comprise a silicon substrate, which may be doped with either a P-type dopant, such as boron, or an N-type dopant, such as phosphorous or arsenic. In addition to silicon, the first semiconductor substrate **100** may also incorporate other elementary semiconductors like germanium. Optionally, it

may include a compound semiconductor or an alloy semiconductor. Moreover, the first semiconductor substrate **100** may feature an epitaxial layer (also known as an epi layer), may be strained to enhance performance, and may include a silicon-on-insulator (SOI) structure.

[0024] The read-out circuits are formed on the front side **102** of the first semiconductor substrate **100**. In some embodiments, the read-out circuits are arranged in rows and columns, and each read-out circuit includes logic devices, such as a transistor **T1** and a transistor **T2**. In some embodiments, the transistor **T1** is a row selector transistor, and the transistor **T2** is a source follower transistor. In some embodiments, each read-out circuit further includes a reset transistor (not shown) electrically connected with the transistor **T2**.

[0025] The channel features **101** and the source/drain features **103** of the transistors **T1** and the transistors **T2** may be formed by performing a plurality of ion implantation processes on the first semiconductor substrate **100**. The ion implantation processes may include multiple implant steps and may use different types of dopants, implant dosages, and implantation energies. The ion implantation processes may also use different masks that have different patterns and opening sizes.

[0026] A gate insulation layer **110** and a gate electrode layer **120** are formed over the front side **102** of the first semiconductor substrate **100**. The gate insulation layer **110** includes gate insulators **112** and an insulation structure **114**. In some embodiments, the gate insulators **112** and the insulation structure **114** are separated from each other, but the disclosure is not limited thereto. In other embodiments, the gate insulators **112** and the insulation structure **114** are connected with each other. In some embodiments, the gate insulation layer **110** comprises silicon oxide. In some embodiments, the gate insulation layer **110** comprises a high-k dielectric material. As used herein, the term “high-k dielectric” refers to the material having a dielectric constant, κ , greater than or equal to about 3.9, which is the k value of SiO₂. The material of the high-k dielectric layer may be any suitable material. Examples of the material of the high-k dielectric layer include but are not limited to Al₂O₃, HfO₂, ZrO₂, La₂O₃, TiO₂, SrTiO₃, LaAlO₃, Y₂O₃, Al₂O_xN_y, HfO_xN_y, ZrO_xN_y, La₂O_xN_y, TiO_xN_y, SrTiO_xN_y, LaAlO_xN_y, Y₂O_xN_y, SiON, SiN_x, a silicate thereof, and an alloy thereof. Each value of x is independently from 0.5 to 3, and each value of y is independently from 0 to 2. In some embodiments, the thickness of the gate insulation layer **110** is in a range between approximately 10 Angstroms to approximately 1000 Angstroms.

[0027] The gate electrode layer **120** includes gate electrodes **122** and a conductive pad **124**. The gate insulators **112** are located between the gate electrodes **122** and the first semiconductor substrate **100**, and the insulation structure **114** is located between the conductive pad **124** and the first semiconductor substrate **100**. In some embodiments, the sidewalls of the gate electrodes **122** are aligned with the sidewalls of the gate insulators **112**, and the sidewalls of the conductive pad **124** are aligned with the sidewalls of the insulation structure **114**, but the disclosure is not limited thereto.

[0028] In some embodiments, the gate electrode layer **120** is made of polysilicon, metal (such as copper, tungsten, or the like) or other conductive materials. In some embodiments, the thickness of the gate electrode layer **120** is in a range between approximately 500 Angstroms to approximately 3000 Angstroms.

[0029] An interlayer dielectric (ILD) structure **130** is disposed over the transistors **T1** and the transistors **T2**. In some embodiments, the ILD structure **130** includes two or more dielectric layers. For example, the ILD structure **130** includes a first ILD layer **132** and a second ILD layer **134**. The first ILD layer **132** is disposed over the transistors **T1** and the transistors **T2**. The second ILD layer **134** is disposed over the first ILD layer **132**.

[0030] In some embodiments, the gate electrodes **122** and the conductive pad **124** are embedded in the first ILD layer **132** and located between the first ILD layer **132** and the first semiconductor substrate **100**. In some embodiments, the first ILD layer **132** may include silicon oxide, SiOC, SiOCN or SiCN or other low-k materials, or porous materials, or any other suitable dielectric

material. In some embodiments, the first ILD layer **132** may include low-k dielectric materials having κ values, for example, lower than about 3.9 or even 2.0. The first ILD layer **132** can be formed by LPCVD (low pressure chemical vapor deposition), plasma-CVD, flowable CVD or other suitable film forming methods.

[0031] In some embodiments, formation of the first ILD layer **132** is followed by a damascene process to form openings within the first ILD layer **132** and fill those openings with a metal material (e.g., copper, aluminum, tungsten, and so on). A planarization process can be then performed to remove excess metal material to form an interlayer conductive layer **138** within the first ILD layer **132**. In some embodiments, the interlayer conductive layer **138** interconnect structure may be referred to as **M0** layer. In some embodiments, the thickness of the interlayer conductive layer **138** is in a range between approximately 50 Angstroms to approximately 3000 Angstroms.

[0032] The second ILD layer **134** is disposed over the first ILD layer **132** and covers the interlayer conductive layer **138**. That is, the interlayer conductive layer **138** is embedded in the ILD structure **130** and located between the first ILD layer **132** and the second ILD layer **134**. In some embodiments, the second ILD layer **134** may include silicon oxide, SiOC, SiOCN or SiCN or other low-k materials, or porous materials, or any other suitable dielectric material. In some embodiments, the second ILD layer **134** may include low-k dielectric materials having k values, for example, lower than about 4.0 or even 2.0. The second ILD layer **134** can be formed by LPCVD (low pressure chemical vapor deposition), plasma-CVD, flowable CVD or other suitable film forming methods.

[0033] In some embodiments, the first ILD layer **132** has a thickness **Z1** in a range between approximately 10 Angstroms to approximately 3000 Angstroms. In some embodiments, the second ILD layer **134** has a thickness **Z2** in a range between approximately 10 Angstroms to approximately 3000 Angstroms.

[0034] The source/drain contact vias **135** and the conductive via **137** are embedded within the ILD structure **130**. In some embodiments, after forming the second ILD layer **134**, one or more etching processes are performed to create multiple openings in the ILD structure **130**. Some of these openings expose parts of the source/drain features **103**, while others expose the conductive pad **124**. A metal material (e.g., copper, aluminum, tungsten, and so on) is then filled into these openings. Subsequently, a planarization process is performed to remove excess metal material, thereby forming the source/drain contact vias **135** and the conductive via **137** within the ILD structure **130**.

[0035] In some embodiments, in addition to forming the source/drain contact vias **135** and the conductive via **137** within the ILD structure **130**, gate contact vias (not shown) are also formed on the gate electrodes **122** within the ILD structure **130**.

[0036] The source/drain contact vias **135** are extending from a top surface of the ILD structure **130** to corresponding source/drain features **103** of the transistors **T1**, **T2**, and electrically connected with the corresponding source/drain features **103** of the transistors **T1**, **T2**. In some embodiments, the source/drain feature **103**, located between one transistor **T1** and an adjacent transistor **T2**, is not connected with any source/drain contact via **135**, but the disclosure is not limited thereto. In some embodiments, the source/drain contact vias **135** are penetrating through the first ILD layer **132** and the second ILD layer **134**.

[0037] The conductive via **137** is extending from the top surface of the ILD structure **130** into the ILD structure **130**. The conductive via **137** is disposed over and electrically connected with the conductive pad **124**. In some embodiments, the conductive via **137** is penetrating through the second ILD layer **134**, but not penetrating through the first ILD layer **132**. In some embodiments, a height **H1** of the source/drain contact vias **135** is greater than a height **H2** of the conductive via **137**.

[0038] An interconnect structure **140** is disposed over the ILD structure **130** and electrically

connected with the conductive via **137** and the source/drain contact vias **135**. In certain embodiments, referring to both FIGS. **1B** and **2A**, multiple transistors **T1** are arranged in the same column and are segmented into several groups. Each group contains one or more transistors **T1**, and the transistors **T1** within each group are electrically interconnected through the interconnect structure **140**. However, the source/drain features **103** of the transistors **T1** in different groups are not electrically interconnected to each other through the interconnect structure **140**. In some embodiments, the transistors **T1** in the same group are electrically connected to the same conductive via **137** through the interconnect structure **140**. For example, one of the source/drain features **103** of a transistor **T1** is electrically connected to the corresponding source/drain feature **103** of an adjacent transistor **T1** through the interconnect structure **140**.

[0039] In this embodiment, the source/drain features **103** of the transistors **T1** are electrically connected to the conductive via **137**, but this disclosure is not limited to this. In other embodiments, the gate electrodes **122** of the transistors **T1** or the transistors **T2** are electrically connected to the conductive via **137**.

[0040] The interconnect structure **140** may include a plurality of metallization layers **142** and a plurality of metal vias **144**. The number of metallization layers **142** may vary according to design specifications of the semiconductor device. The metallization layers **142** closest to the ILD structure **130** can be referred to as the **M1** layer. The **M1** layer is in contact with the source/drain contact vias **135** and the conductive via **137**. The metallization layers **142** at different levels are electrically interconnected through metal vias **144**. The metallization layers **142** and the metal vias **144** are located within the inter-metallization dielectric (IMD) layers **146**.

[0041] In some embodiments, the IMD layers **146** may, for example, be or comprise an oxide (e.g., SiO₂), a low-k dielectric, an extreme low-k dielectric, or a combination of the aforementioned dielectric materials formed within a thickness of approximately 200 Angstroms to approximately 9000 Angstroms. As used herein, a low-k dielectric may be, for example, a dielectric with a dielectric constant κ lower than about 3.9 or even 2.0. In some embodiments, etch stop layers (not shown) are located between the adjacent IMD layers **146**. The etch stop layers may, for example, be or comprise silicon carbide (SiC), silicon nitride (SiN), silicon oxynitride (SiON), silicon oxycarbide (SiOC), an oxide layer, or a combination of the aforementioned dielectric materials.

[0042] The topmost layer of the interconnect structure **140** includes bonding pads **148** and a bonding layer **147**. Both the bonding pads **148** and the bonding layer **147** are used in the hybrid bonding process. Specifically, the bonding pads **148** are used for metal-to-metal bonding, while the bonding layer **147** is used for insulator-to-insulator (or dielectric-to-dielectric) bonding during the bonding process.

[0043] A buffer layer **150** is formed on the back side **104** of the first semiconductor substrate **100**. In some embodiments, the buffer layer **150** comprises oxide (such as silicon oxide) or other dielectric materials. In certain embodiments, the buffer layer **150** is utilized to protect the back side **104** of the first semiconductor substrate **100** during subsequent chemical mechanical planarization (CMP) or etching processes.

[0044] Referring to FIG. **2B**, a through hole TH is formed in the first semiconductor substrate **100** by a wet etching process, a dry etching process, or a combination thereof. In some embodiments, the through hole TH passes through both the buffer layer **150** and the first semiconductor substrate **100**, stopping at the gate insulation layer **110**. This arrangement helps to prevent the etching process from damaging the conductive pad **124**. If the etching process does damage the conductive pad **124**, the ions from the conductive pad **124** could potentially contaminate the first semiconductor substrate **100**. This is especially the case during the formation of the through hole TH using a dry etching process. If the dry etching process were to penetrate the gate insulation layer **110**, it could potentially result in damage to the conductive pad **124**.

[0045] However, this disclosure is not limited to having the through hole TH stop at the gate insulation layer **110**. In some embodiments, by controlling the etching process, the through hole TH

can pass through the gate insulation layer **110** and stop at the conductive pad **124**. The aforementioned etching process can be a combination of the wet etching process and the dry etching process.

[0046] Referring to FIG. 2C, a blanket dielectric layer **160** is formed over the back side **104** of the first semiconductor substrate **100**. The blanket dielectric layer **160** is formed on the buffer layer **150** and in the through hole TH. The blanket dielectric layer **160** covers the sidewalls of the through hole TH and the gate insulation layer **110** exposed by the through hole TH. The blanket dielectric layer **160** is in contact with the gate insulation layer **110**. In some embodiments, the through hole TH extends through the gate insulation layer **110**, exposing the conductive pad **124**. In such cases, the blanket dielectric layer **160** comes into contact with the conductive pad **124**. In some embodiments, the blanket dielectric layer **160** includes oxides (such as silicon oxide) or other dielectric materials. In some embodiments, the methods for forming the blanket dielectric layer **160** include physical vapor deposition (PVD), chemical vapor deposition (CVD), atomic layer deposition (ALD), or the like.

[0047] A mask layer **170** is disposed above the blanket dielectric layer **160**. The mask layer **170** is formed over the back side **104** of the first semiconductor substrate **100**. In some embodiments, by adjusting the deposition rate of the mask layer **170**, it is ensured that the mask layer **170** does not close the through hole TH. Therefore, there is no need to use a photomask process to pattern the mask layer **170**. In some embodiments, the deposition rate of the mask layer **170** is greater than that of the blanket dielectric layer **160**. In some embodiments, the material of the mask layer **170** includes nitrides (such as silicon nitride) or other dielectric materials.

[0048] In some embodiments, a part of the mask layer **170** is filled in the through hole TH, but does not completely cover the blanket dielectric layer **160** within the through hole TH. In other words, the part of the mask layer **170** is filled into the through hole TH and has an opening that exposes the blanket dielectric layer **160** in the through hole TH. The width of the opening in mask layer **170** is smaller than that of the through hole TH.

[0049] Referring to FIG. 2D, an etching process is performed on the blanket dielectric layer **160** and the gate insulation layer **110** to form a through hole TH1 passing through the blanket dielectric layer **160** and the gate insulation layer **110**. In some embodiments, the width of the through hole TH is greater than that of the through hole TH1.

[0050] In some embodiments, the mask layer **170** is used as a mask to perform an etching process. This process etches the blanket dielectric layer **160** and the gate insulation layer **110** located at the bottom of the through hole TH, thereby exposing the conductive pad **124**. In some embodiments, the blanket dielectric layer **160** extends through the gate insulation layer **110** and contacts the conductive pad **124**. In such cases, the aforementioned etching process only needs to etch the blanket dielectric layer **160** in the through hole TH to expose the conductive pad **124**.

[0051] Referring to FIG. 2E, a barrier material layer **182** is formed over the mask layer **170** and fills into the through holes TH, TH1. In some embodiments, the barrier material layer **182** comes into contact with the blanket dielectric layer **160** within the through holes TH, and contacts the sidewalls of the gate insulation layer **110** within the through holes TH1.

[0052] A metal layer **184** is then deposited over the barrier material layer **182** and into the through hole TH (or the through holes TH, TH1). In some embodiments, the metal layer **184** may consist of copper or copper alloys, and the barrier material layer **182** may include a nitride (such as titanium nitride, tantalum nitride, or the like) to prevent, for example, the diffusion of copper ions. In other embodiments, the metal layer **184** may comprise other metal materials (such as tungsten), and the barrier material layer **182** can be omitted.

[0053] Referring to FIG. 2F, a CMP process is performed to remove excess materials from the metal layer **184** and the barrier layer **182**, thereby forming the conductive plug **184'** and the barrier layer **182'**. The conductive plug **184'** is embedded within the first semiconductor substrate **100**, with the blanket dielectric layer **160** and the barrier layer **182'** located between the conductive plug

184' and the first semiconductor sub strate **100**. The barrier layer **182'** surrounds the conductive plug **184'**. In some embodiments, a portion of the mask layer **170** is laterally located between the barrier layer **182'** and the blanket dielectric layer **160**.

[0054] In this embodiment, the CMP process stops at the mask layer **170**. The portions of the metal layer **184** and the barrier material layer **182** that extend beyond the mask layer **170** are removed during the CMP process. However, this disclosure is not limited thereto. In other embodiments, the CMP process stops at the blanket dielectric layer **160** or the buffer layer **150**, and any materials extending beyond the blanket dielectric layer **160** or the buffer layer **150** are removed during the CMP process.

[0055] In this embodiment, a TSV **12** includes the barrier layer **182'** and the conductive plug **184'**, extending from the back side **104** of the first semiconductor substrate **100** to beyond the front side **102** of the first semiconductor substrate **100**. In this instance, the TSV **12** penetrates through the insulation structure **114** of the gate insulation layer **110** and is in contact with the conductive pad **124**. In some embodiments, a depth **Y1** of the TSV **12** beyond the front side **102** of the first semiconductor substrate **100** is less than or equal to a thickness **Z1** of the first ILD layer **132**. In some instances, the depth **Y1** is approximately equal to the thickness of the gate insulation layer **110**.

[0056] The conductive pad **124** is located between the TSV **12** and the conductive via **137**. The TSV **12** is electrically connected with the conductive via **137** through the conductive pad **124**, and further electrically connected with the transistor **T1** through the interconnect structure **140** and the source/drain contact via **135**.

[0057] In this embodiment, since the TSV **12** does not extend through the ILD structure **130**, the resulting TSV **12** may have a smaller depth, for example, ranging from 0.1 micrometers to 3 micrometers. Due to the reduction in the depth of the TSV **12**, the TSV **12** may have a smaller critical dimension (CD) **W1**, for instance, ranging from 0.05 micrometers to 3 micrometers.

[0058] Then, a backside circuit is formed over the back side **104** of the first semiconductor substrate **100**, as shown in FIGS. 2G to 2I. Referring to FIG. 2G, a nitride layer **192** is formed over the back side **104** and covers the TSV **12**. In this embodiment, since the CMP process of forming the TSV **12** stops on the mask layer **170**, the nitride layer **192** is formed on the mask layer **170**. In other embodiments, the aforementioned CMP process stops on the blanket dielectric layer **160** or the buffer layer **150**, and the nitride layer **192** is formed on the blanket dielectric layer **160** or the buffer layer **150**. In some embodiments, the nitride layer **192** includes silicon nitride or other suitable materials. A dielectric material layer **194** is formed over the nitride layer **192**. In some embodiments, the dielectric material layer **194** includes silicon oxide or other suitable materials.

[0059] Referring to FIG. 2H, a patterning process is performed on the dielectric material layer **194** and the nitride layer **192** to form a dielectric layer **194'** and a nitride layer **192'** with openings and/or trenches. The TSV **12** is exposed after the patterning process.

[0060] Referring to FIG. 2I, a metal layer **196** is formed in the openings and/or trenches of the dielectric layer **194'** and the nitride layer **192'**. In some embodiments, a metal material is first deposited over the dielectric layer **194'**, followed by a CMP process to remove the metal material exceeding the dielectric layer **194'**, leaving behind the metal layer **196**.

[0061] In this embodiment, the backside circuit includes the metal layer **196**, but this disclosure is not limited to this. The backside circuit may also include multiple layers of metal layers. Furthermore, in some embodiments, the backside circuit may also include bonding pads and/or a bonding layer for bonding with other devices.

[0062] Referring to FIG. 2J, a second semiconductor structure **20** is bonding with the first semiconductor structure **10** (as shown in FIG. 21) by hybrid bonding process.

[0063] The second semiconductor structure **20** includes a second semiconductor substrate **200**, an ILD structure **230**, and an interconnect structure **240**.

[0064] The second semiconductor substrate **200** may comprise a silicon substrate, which may be

doped with either a P-type dopant, such as boron, or an N-type dopant, such as phosphorous or arsenic. In addition to silicon, the second semiconductor substrate **200** may also incorporate other elementary semiconductors like germanium. Optionally, it may include a compound semiconductor or an alloy semiconductor. Moreover, the second semiconductor substrate **200** may feature an epitaxial layer (also known as an epi layer), may be strained to enhance performance, and may include a silicon-on-insulator (SOI) structure.

[0065] The image sensor IS includes a radiation-sensing region **201**, a transfer gate **220**, and a charge-to-voltage conversion region **205**. The radiation-sensing region **201** is located in the second semiconductor substrate **200**, adjacent to the transfer gate **220**. This region is designed to receive and accumulate radiation waves entering from the back side of the second semiconductor substrate **200**. The radiation-sensing region **201**, which can be a photodiode, a pinned photodiode, or a p-n junction located in the second semiconductor substrate **200**, is capable of generating a signal that corresponds to the intensity or brightness of the light striking the radiation-sensing region **201**.

[0066] The image sensors IS are arrayed in a pixel area of the second semiconductor substrate **20** and arranged in multiple rows and columns, as shown in FIG. **1C**. In other words, the radiation-sensing region **201**, the transfer gate **220**, and the charge-to-voltage conversion region **205** are disposed in the pixel area.

[0067] The transfer gate **220** is located laterally adjacent to and coupled with the radiation-sensing region **201**. The transfer gate **220** is located between the radiation-sensing region **201** and the charge-to-voltage conversion region **205**. The transfer gate **220**, which can be a control gate with a metal gate structure or a polysilicon structure, controls the transfer of electrons or charges from the radiation-sensing region **201** to the charge-to-voltage conversion region **205**. Furthermore, the charge-to-voltage conversion region **205** functions as a region designed to convert the electrons or charges into a corresponding signal. The gate insulation layer **210** is located between the transfer gate **220** and the second semiconductor substrate **200**.

[0068] The ILD structure **230** is disposed over the transfer gates **220**. Gate contact vias **232** are embedded in the ILD structure **230** and electrically connected with the transfer gates **220**.

[0069] The interconnect structure **240** is disposed over the ILD structure **230** and electrically connected with the gate contact vias **232**. The interconnect structure **240** may include a plurality of metallization layers **242** and a plurality of metal vias **244**. The number of metallization layers **242** may vary according to design specifications of the semiconductor device. The metallization layers **242** at different levels are electrically interconnected through metal vias **244**. The metallization layers **242** and the metal vias **244** are located within the IMD layers **246**.

[0070] The topmost layer of the interconnect structure **240** includes bonding pads **248** and a bonding layer **247**. Both the bonding pads **248** and the bonding layer **247** are used in the hybrid bonding process. Specifically, the bonding pads **248** are bonded with the bonding pads **148**, while the bonding layer **247** is bonded with the bonding layer **147** during the bonding process.

[0071] The image sensors IS are electrically connected to the transistors **T1**, **T2** via the interconnect structures **140**, **240**. In some embodiments, reducing the distance between the TSV **12** and the transistor **T1** may reduce the signal transmission path between the image sensor IS and the corresponding TSV **12**, thereby improving the RC delay problem. In some embodiments, the TSV **12** is overlapping with the pixel area composed of the array of the image sensors IS. For example, the TSV **12** is overlapping with the second semiconductor substrate **200** between two adjacent image sensors IS or overlapping with the image sensor IS.

[0072] In some embodiments, the image sensor device **2** includes the second semiconductor structure **20** and the first semiconductor structure **10**, but this disclosure is not limited thereto. In other embodiments, the image sensor device **2** further includes other semiconductor structures, such as ASIC, bonded to the back side of the first semiconductor structure **10**. In some embodiments, optical elements such as color filters, lens, or other optical elements may be placed on the back side of the second semiconductor structure **20** facing away from the first

semiconductor structure **10**, but this disclosure is not limited thereto.

[0073] FIG. **3** is a schematic cross-sectional view of an embodiment first semiconductor structure **10A** in accordance with various embodiments. The first semiconductor structure **10A** in FIG. **3** is similar to the first semiconductor structure **10** in FIG. **2I**. The differences between the two are explained below.

[0074] In the first semiconductor structure **10** of FIG. **2I**, the etching process for forming the through hole TH in the first semiconductor substrate **100** stops on the insulation structure **114** of the gate insulation layer **110** (as shown in FIG. **2B**), so that the subsequently formed the blanket dielectric layer **160** contacts the bottom surface **114b** of the insulation structure **114** (as shown in FIG. **2C**). However, in the first semiconductor structure **10A** of FIG. **3**, the etching process for forming the through hole TH stops on the conductive pad **124** of the gate electrode layer **120**, so that the aforementioned etching process forms an opening in the insulation structure **114**, and the subsequently formed blanket dielectric layer **160** contacts the sidewall of the aforementioned opening (which may be referred to as the sidewall of the insulation structure **114**). Therefore, in the process of the first semiconductor structure **10A**, the step of etching the blanket dielectric layer **160** (similar to the step of FIG. **2D**) may expose the conductive pad **124** without etching the insulation structure **114**.

[0075] In addition, in the first semiconductor structure **10A** of FIG. **3**, the CMP process for grinding the metal material in the through hole TH (similar to the step of FIG. **2F**) stops on the buffer layer **150**, and the materials beyond the buffer layer **150** (including the metal layer **184** and the barrier layer **182**, the blanket dielectric layer **160** and the mask layer **170** above the buffer layer **150** shown in FIG. **2E**) are removed in the CMP process. In the first semiconductor structure **10A** of FIG. **3**, a portion of the blanket dielectric layer beyond the buffer layer **150** is removed during the CMP process, remaining the dielectric layer **160'**. In some embodiments, a portion of the mask layer beyond the buffer layer **150** is removed during the CMP process, remaining the mask layer **170'**. In some embodiments, the mask layer **170'** is laterally disposed between the barrier layer **182'** and the dielectric layer **160'**. A portion of the barrier layer **182'** is in contact with the dielectric layer **160'**, and another portion of the barrier layer **182'** is in contact with the mask layer **170'**.

[0076] In some implementations, the first semiconductor structure **10A** and the second semiconductor structure **20** are bonded together using a method similar to that disclosed in FIG. **2J** to obtain an image sensor device.

[0077] FIGS. **4A** to **4H** are schematic cross-sectional views of a fabrication method of an embodiment image sensor device **3** in accordance with various embodiments. The structure in FIG. **4A** is similar to the structure in FIG. **2A**, with the following differences: In the structure of FIG. **4A**, the gate insulation layer **110** does not include the insulation structure **114** (as shown in FIG. **2A**), the gate electrode layer **120** does not include conductive pad **124** (as shown in FIG. **2A**), and the interlayer conductive layer **138** includes a conductive pad **138A**.

[0078] Referring to FIG. **4A**, the conductive pad **138A** is embedded in the ILD structure **130** and located between the first ILD layer **132** and the second ILD layer **134**. A portion of the ILD structure **130** (such as a portion of the first ILD layer **132**) is located between the conductive pad **138A** and the first semiconductor substrate **100**. The conductive via **137** is embedded in the ILD structure **130** and disposed over the conductive pad **138A**. In some embodiments, the source/drain contact vias **135** is penetrating through the first ILD layer **132** and the second ILD layer **134**. In some embodiments, the conductive via **137** is penetrating through the second ILD layer **134**, but not penetrating through the first ILD layer **132**. A height H1 of the source/drain contact via **135** is greater than a height H2 of the conductive via **137**.

[0079] Referring to FIG. **4B**, a through hole TH is formed in the first semiconductor substrate **100** by a wet etching process, a dry etching process, or a combination thereof. In some embodiments, the through hole TH passes through both the buffer layer **150**, the first semiconductor substrate **100**, and the first ILD layer **132**, stopping at the conductive pad **138A**. In this embodiment, the first

ILD layer **132** is located between the first semiconductor substrate **100** and the conductive pad **138A**, which helps to reduce the contamination of the first semiconductor substrate **100** by ions from the conductive pad **138A** during the formation of the through hole TH.

[0080] Referring to FIG. 4C, a blanket dielectric layer **160** is formed over the back side **104** of the first semiconductor substrate **100**. The blanket dielectric layer **160** is formed on the buffer layer **150** and in the through hole TH. The blanket dielectric layer **160** covers the sidewalls of the through hole TH and the conductive pad **138A** exposed by the through hole TH. The blanket dielectric layer **160** is in contact with the conductive pad **138A**.

[0081] A mask layer **170** is disposed above the blanket dielectric layer **160**. The mask layer **170** is formed over the back side **104** of the first semiconductor substrate **100**. In some embodiments, part of the mask layer **170** is filled in the through hole TH, but does not completely cover the blanket dielectric layer **160** within the through hole TH. In other words, the part of the mask layer **170** is filled into the through hole TH and has an opening that exposes the blanket dielectric layer **160** in the through hole TH. The width of the opening in mask layer **170** is smaller than that of the through hole TH.

[0082] Referring to FIG. 4D, an etching process is performed on the blanket dielectric layer **160** to form a through hole TH1 passing through the blanket dielectric layer **160**. In some embodiments, the width of the through hole TH is greater than that of the through hole TH1.

[0083] In some embodiments, the mask layer **170** is used as a mask to perform an etching process. This process etches the blanket dielectric layer **160** located at the bottom of the through hole TH, thereby exposing the conductive pad **138A**.

[0084] Referring to FIG. 4E, a barrier material layer **182** is formed over the mask layer **170** and fills into the through holes TH, TH1. In some embodiments, the barrier material layer **182** comes into contact with the blanket dielectric layer **160**, the sidewalls of the through holes TH1, and the conductive pad **138A**.

[0085] A metal layer **184** is then deposited over the barrier material layer **182** and into the through hole TH (or the through holes TH, TH1). In some embodiments, the metal layer **184** may consist of copper or copper alloys, and the barrier material layer **182** may include a nitride (such as titanium nitride, tantalum nitride, or the like) to prevent, for example, the diffusion of copper ions. In other embodiments, the metal layer **184** may comprise other metal materials (such as tungsten), and the barrier material layer **182** can be omitted. In some embodiments, the metal layer **184** and the conductive pad **138A** includes the same materials.

[0086] Referring to FIG. 4F, a CMP process is performed to remove excess materials from the metal layer **184** and the barrier layer **182**, thereby forming the conductive plug **184'** and the barrier layer **182'**. The conductive plug **184'** is embedded within the first semiconductor substrate **100**. In this embodiment, the CMP process stops at the buffer layer **150**, and any materials extending beyond the buffer layer **150** are removed during the CMP process. In some embodiments, a portion of the blanket dielectric layer **160** beyond the buffer layer **150** are removed during the CMP process, remaining the dielectric layer **160'**. In some embodiments, a portion of the mask layer **170** beyond the buffer layer **150** are removed during the CMP process, remaining the mask layer **170'**.

[0087] In this embodiment, a TSV **12** includes the dielectric layer **160'**, the barrier layer **182'**, and the conductive plug **184'**. The TSV **12** is extending from the back side **104** of the first semiconductor substrate **100** to beyond the front side **102** of the first semiconductor substrate **100**. In this instance, the TSV **12** penetrates through the first ILD layer **132** and is in contact with the conductive pad **138A**. In some embodiments, a depth Y1 of the TSV **12** beyond the front side **102** of the first semiconductor substrate **100** is less than or equal to a thickness Z1 of the first ILD layer **132**.

[0088] The TSV **12** is electrically connected with the conductive via **137** through the conductive pad **138A**, and further electrically connected with the transistor T1 through the interconnect structure **140** and the source/drain contact via **135**.

[0089] In this embodiment, since the TSV **12** does not extend through the ILD structure **130**, the resulting TSV **12** may have a smaller depth, for example, ranging from 0.1 micrometers to 3 micrometers. Due to the reduction in the depth of the TSV **12**, the TSV **12** may have a smaller critical dimension (CD), for instance, ranging from 0.05 micrometers to 3 micrometers.

[0090] Then, a backside circuit is formed on the back side **104** of the first semiconductor substrate **100**, as shown in FIG. **4G**. The process for forming a backside circuit can be referred to in FIGS. **2G** to **2I** and their related descriptions.

[0091] Finally, the first semiconductor substrate **10B** shown in FIG. **4G** is bonded with the second semiconductor **20** as shown in FIG. **4H**. The bonding process can be referred to in FIG. **2J** and the related descriptions.

[0092] The image sensors **IS** are electrically connected to the transistors **T1**, **T2** via the interconnect structures **140**, **240**. In some embodiments, reducing the distance between the TSV **12** and the transistor **T1** may reduce the signal transmission path between the image sensor **IS** and the corresponding TSV **12**, thereby improving the RC delay problem. In some embodiments, the TSV **12** is overlapping with the pixel area composed of an array of the image sensors **IS**. For example, the TSV **12** is overlapping with the second semiconductor substrate **200** between two adjacent image sensors **IS** or overlapping with the image sensor **IS**.

[0093] In some embodiments, the image sensor device **3** includes the second semiconductor structure **20** and the first semiconductor structure **10B**, but this disclosure is not limited thereto. In other embodiments, the image sensor device **3** further includes other semiconductor structures, such as ASIC, bonded to the back side of the first semiconductor structure **10B**. In some embodiments, optical elements such as color filters, lens, or other optical elements may be placed on the back side of the second semiconductor structure **20** facing away from the first semiconductor structure **10B**, but this disclosure is not limited thereto.

[0094] FIGS. **5A** to **5N** are schematic cross-sectional views of a fabrication method of an embodiment image sensor device **4** in accordance with various embodiments. Referring to FIG. **5A**, a first semiconductor substrate **100** is provided. The read-out circuits are formed on the front side **102** of the first semiconductor substrate **100**. In some embodiments, the read-out circuits are arranged in rows and columns, and each read-out circuit includes logic devices, such as a transistor **T1** and a transistor **T2**. In some embodiments, the transistor **T1** is a row selector transistor, and the transistor **T2** is a source follower transistor. In some embodiments, each read-out circuit further includes a reset transistor (not shown) electrically connected with the transistor **T2**.

[0095] The channel features **101** and the source/drain features **103** of the transistors **T1** and the transistors **T2** may be formed by performing a plurality of ion implantation processes on the first semiconductor substrate **100**. The ion implantation processes may include multiple implant steps and may use different types of dopants, implant dosages, and implantation energies. The ion implantation processes may also use different masks that have different patterns and opening sizes.

[0096] A gate insulation layer **110** and the gate electrode layer **120** are formed over the front side **102** of the first semiconductor substrate **100**. The gate insulation layer **110** includes gate insulators **112**. In some embodiments, the gate insulators **112** are separated from each other, but the disclosure is not limited thereto. In other embodiments, the gate insulators **112** are connected with each other. In some embodiments, the thickness of the gate insulation layer **110** is in a range between approximately 10 Angstroms to approximately 1000 Angstroms.

[0097] The gate electrode layer **120** includes gate electrodes **122**. The gate insulators **112** are located between the gate electrodes **122** and the first semiconductor substrate **100**. In some embodiments, the sidewalls of the gate electrodes **122** are aligned with the sidewalls of the gate insulators **112**, but the disclosure is not limited thereto. In some embodiments, the thickness of the gate electrode layer **120** is in a range between approximately 500 Angstroms to approximately 3000 Angstroms.

[0098] A first ILD layer **132** is disposed over the front side **102** of the first semiconductor substrate

100 and covers the transistors **T1** and the transistors **T2**.

[0099] Referring to FIG. 5B, one or more etching processes are performed to form an opening **O1** in the first ILD layer **132** and the first semiconductor substrate **100**. In some embodiments, a patterned photoresist layer (not shown) is formed over the first ILD layer **132**. Wet etching process and/or dry etching process is then performed using the patterned photoresist layer as a mask to form the opening **O1**. In some embodiments, the opening **O1** extends through the first ILD layer **132** and into the first semiconductor substrate **100** from the front side **102** of the first semiconductor substrate **100**, but not extending to the back side **104** of the first semiconductor substrate **100**.

[0100] In some embodiments, if the depth of the opening **O1** is deeper, the width of the opening **O1** may be wider under a fixed aspect ratio. In some embodiments, since the opening **O1** does not need to penetrate the first semiconductor substrate **100**, the width of the opening **O1** may be reduced by making the depth of the opening **O1** shallower. In some embodiments, the depth **X1** of the opening **O1** is ranging from 0.01 micrometers to 5 micrometers, and a width **W2** of the opening **O1** is ranging from 0.01 micrometers to 5 micrometers.

[0101] Referring to FIG. 5C, a first blanket dielectric layer **131A** is formed over the first ILD layer **132** and filled into the opening **O1**. The first blanket dielectric layer **131A** covers the sidewalls and the bottom surface of the opening **O1**. In some embodiments, the first blanket dielectric layer **131A** includes oxides (such as silicon oxide) or other dielectric materials. In some embodiments, the methods for forming the first blanket dielectric layer **131A** include PVD, CVD, ALD, or the like.

[0102] A first barrier material layer **131B** is formed over the first blanket dielectric layer **131A** and fills into the opening **O1**. In some embodiments, the first barrier material layer **131B** is in contact with the first blanket dielectric layer **131A**.

[0103] A first metal layer **131C** is then deposited over the first barrier material layer **131B** and into the opening **O1**. In some embodiments, the first metal layer **131C** may consist of copper or copper alloys, and the first barrier material layer **131B** may include a nitride (such as titanium nitride, tantalum nitride, or the like) to prevent, for example, the diffusion of copper ions. In other embodiments, the first metal layer **131C** may comprise other metal materials (such as tungsten), and the first barrier material layer **131B** can be omitted.

[0104] Referring to FIG. 5D, a CMP process is performed to remove excess materials from the first metal layer **131C**, the first barrier material layer **131B**, and the first blanket dielectric layer **131A**, thereby forming the first conductive plug **131C'**, the first barrier layer **131B'**, and the first dielectric layer **131A'**. The first dielectric layer **131A'** is laterally disposed between the first conductive plug **131C'** and the first semiconductor substrate **100**. The first barrier layer **131B'** is laterally disposed between the first conductive plug **131C'** and the first dielectric layer **131A'**.

[0105] In this embodiment, the CMP process stops at the first ILD layer **132**. The portions of the first metal layer **131C**, the first barrier material layer **131B**, and the first blanket dielectric layer **131A** that extend beyond the first ILD layer **132** are removed during the CMP process. In other embodiments, the CMP process stops at the first blanket dielectric layer **131A**.

[0106] The first conductive plug **131C'**, the first barrier layer **131B'**, and the first dielectric layer **131A'** extend from the top surface of the first ILD layer **132** to the front side **102** of the first semiconductor substrate **100**, and into the first semiconductor substrate **100** from the front side **102**.

[0107] Referring to FIG. 5E, the second ILD layer **134** is formed over the first ILD layer **132** and covers the first conductive plug **131C'**, the first barrier layer **131B'**, and the first dielectric layer **131A'**. That is, the first conductive plug **131C'**, the first barrier layer **131B'**, and the first dielectric layer **131A'** are embedded in the ILD structure **130**. The first conductive plug **131C'** is extending from the ILD structure **130** into the first semiconductor substrate **100**.

[0108] In some embodiments, the first ILD layer **132** has a thickness **Z1** in a range between approximately 10 Angstroms to approximately 3000 Angstroms. In some embodiments, the second

ILD layer **134** has a thickness **Z2** in a range between approximately 10 Angstroms to approximately 3000 Angstroms.

[0109] Referring to FIG. 5F, source/drain contact vias **135** and the conductive via **137** are formed within the ILD structure **130**. In some embodiments, after forming the second ILD layer **134**, one or more etching processes are performed to create multiple openings in the ILD structure **130**. Some of these openings expose parts of the source/drain features **103**, while others expose the first conductive plug **131C'**. A metal material (e.g., copper, aluminum, tungsten, and so on) is then filled into these openings. Subsequently, a planarization process is performed to remove excess metal material, thereby forming the source/drain contact vias **135** and the conductive via **137** within the ILD structure **130**.

[0110] In some embodiments, in addition to forming the source/drain contact vias **135** and the conductive via **137** within the ILD structure **130**, gate contact vias (not shown) are also formed over the gate electrodes **122** within the ILD structure **130**.

[0111] The source/drain contact vias **135** are extending from a top surface of the ILD structure **130** to corresponding source/drain features **103** of the transistors **T1**, **T2**, and electrically connected with the corresponding source/drain features **103** of the transistors **T1**, **T2**. In some embodiments, the source/drain feature **103**, located between one transistor **T1** and an adjacent transistor **T2**, is not connected with the source/drain contact vias **135**, but the disclosure is not limited thereto. In some embodiments, the source/drain contact vias **135** is penetrating through the first ILD layer **132** and the second ILD layer **134**.

[0112] The conductive via **137** are extending from the top surface of the ILD structure **130** into the ILD structure **130**. The conductive via **137** is disposed over and electrically connected with the first conductive plug **131C'**. The conductive via **137** is landed on the first conductive plug **131C'**. In some embodiments, the conductive via **137** is penetrating through the second ILD layer **134**, but not penetrating through the first ILD layer **132**. In some embodiments, a height **H1** of the source/drain contact via **135** is greater than a height **H2** of the conductive via **137**.

[0113] Referring to FIG. 5G, an interconnect structure **140** is disposed over the ILD structure **130** and electrically connected with the conductive via **137** and the source/drain contact vias **135**. In certain embodiments, multiple transistors **T1** are arranged in the same column and are divided into several groups. Each group contains one or more transistors **T1**, and the transistors **T1** within each group are electrically interconnected through the interconnect structure **140**. In some embodiments, the transistors **T1** in the same group are electrically connected to the same conductive via **137** through the interconnect structure **140**. For example, one of the source/drain features **103** of a transistor **T1** is electrically connected to the corresponding source/drain feature **103** of an adjacent transistor **T1** through the interconnect structure **140**. Furthermore, in this embodiment, the source/drain features **103** of the transistors **T1** are electrically connected to the conductive via **137**, but this disclosure is not limited to this. In other embodiments, the gate electrodes **122** of the transistors **T1** or the transistors **T2** are electrically connected to the conductive vias **137**.

[0114] The interconnect structure **140** may include a plurality of metallization layers **142** and a plurality of metal vias **144**. The number of metallization layers **142** may vary according to design specifications of the semiconductor device. The metallization layers **142** closest to the ILD structure **130** can be referred to as the **M1** layer. The **M1** layer is in contact with the source/drain contact vias **135** and the conductive via **137**. The metallization layers **142** at different levels are electrically interconnected through the metal vias **144**. The metallization layers **142** and the metal vias **144** are located within the inter-metallization dielectric (IMD) layers **146**.

[0115] The topmost layer of the interconnect structure **140** includes bonding pads **148** and a bonding layer **147**. Both the bonding pads **148** and the bonding layer **147** are used in the hybrid bonding process. Specifically, the bonding pads **148** are used for metal-to-metal bonding, while the bonding layer **147** is used for insulator-to-insulator (or dielectric-to-dielectric) bonding during the bonding process.

[0116] A buffer layer **150** is formed on the back side **104** of the first semiconductor substrate **100**. In some embodiments, the buffer layer **150** comprises oxide (such as silicon oxide) or other dielectric materials. In certain embodiments, the buffer layer **150** is utilized to protect the back side **104** of the first semiconductor substrate **100** during subsequent chemical mechanical planarization (CMP) or etching processes.

[0117] Referring to FIG. 5H, an opening **O2** is formed in the first semiconductor substrate **100** by a wet etching process, a dry etching process, or a combination thereof. In some embodiments, the opening **O2** passes through the buffer layer **150** and extending into the first semiconductor substrate **100** from the back side **102**, and stopping at the first barrier layer **131B'** and/or the first conductive plug **131C'**.

[0118] In some embodiments, a recess is formed on the back side **102** of the first semiconductor substrate **100** using dry etching. The recess is overlapping with the first conductive plug **131C'**, but does not extend through the first barrier layer **131B'** and the first dielectric layer **131A'**, to prevent damage to the first conductive plug **131C'** caused by the dry etching process and contamination of the first semiconductor substrate **100**. Subsequently, the first dielectric layer **131A'** at the bottom of the aforementioned recess is removed using wet etching, exposing the first barrier layer **131B'** and/or the first conductive plug **131C'**.

[0119] Referring to FIG. 5I, a blanket dielectric layer **161** (which is referred to as the second dielectric layer in some embodiments) is formed over the back side **104** of the first semiconductor substrate **100**. The blanket dielectric layer **161** is formed on the buffer layer **150** and in the opening **O2**. The blanket dielectric layer **161** covers the sidewalls of the opening **O2** and is in contact with the first dielectric layer **131A'**. The blanket dielectric layer **161** also covers the first barrier layer **131B'** and/or the first conductive plug **131C'** that is exposed by the opening **O2**.

[0120] In some embodiments, the blanket dielectric layer **161** includes oxides (such as silicon oxide) or other dielectric materials. In some embodiments, the methods for forming the blanket dielectric layer **161** include PVD, CVD, ALD, or the like.

[0121] A mask layer **171** is disposed above the blanket dielectric layer **161**. The mask layer **171** is formed over the back side **104** of the first semiconductor substrate **100**. In some embodiments, by adjusting the deposition rate of the mask layer **171**, it is ensured that the mask layer **171** does not fill into the opening **O2**. Therefore, there is no need to use a photomask process to pattern the mask layer **170**. In some embodiments, the material of the mask layer **171** includes nitrides (such as silicon nitride) or other dielectric materials.

[0122] In some embodiments, part of the mask layer **171** is filled in the opening **O2**, but does not completely cover the blanket dielectric layer **161** within the opening **O2**. In other words, the part of the mask layer **171** is filled into the opening **O2** and has an opening that exposes the blanket dielectric layer **161** in the opening **O2**. The width of the opening in mask layer **171** is smaller than that of the opening **O2**.

[0123] Referring to FIG. 5J, an etching process is performed on the blanket dielectric layer **161** to form a through hole **O2a** passing through the blanket dielectric layer **161**. In some embodiments, the width of the opening **O2** is greater than that of the through hole **O2a**.

[0124] In some embodiments, the mask layer **171** is used as a mask to perform an etching process. This process etches the blanket dielectric layer **161** located at the bottom of the opening **O2**, thereby exposing the first barrier layer **131B'** and/or the first conductive plug **131C'**.

[0125] In some embodiments, the blanket dielectric layer **161** is passing through the first dielectric layer **131A'** and in contact with the first barrier layer **131B'** and/or the first conductive plug **131C'**, but the disclosure is not limited thereto. In other embodiments, the opening **O2** does not extend through the first dielectric layer **131A'**, and the blanket dielectric layer **161** is separated from the first barrier layer **131B'** by the first dielectric layer **131A'**. In this scenario, the etching process is used to remove the blanket dielectric layer **161** and the first dielectric layer **131A'** at the bottom of the opening **O2** to form the opening exposing the first barrier layer **131B'** and/or the first

conductive plug **131C'**.

[0126] Referring to FIG. 5K, a barrier material layer **186** and a metal layer **188** are formed above the mask layer **171**. In some embodiments, the metal layer **188** may consist of copper or copper alloys, and the barrier material layer **186** may include a nitride (such as titanium nitride, tantalum nitride, or the like) to prevent, for example, the diffusion of copper ions. In other embodiments, the metal layer **188** may comprise other metal materials (such as tungsten), and the barrier material layer **186** can be omitted.

[0127] Referring to FIG. 5L, a CMP process is used to remove the excess portions of the barrier material layer **186** and the metal layer **188**, thereby forming the second barrier layer **186'** and the second conductive plug **188'**. In this embodiment, the CMP process stops at the mask layer **171**. However, this disclosure is not limited thereto. In other embodiments, the CMP process stops at the blanket dielectric layer **161** or the buffer layer **150**, and any materials extending beyond the blanket dielectric layer **161** or the buffer layer **150** are removed during the CMP process.

[0128] The second conductive plug **188'** and the second barrier layer **186'** are formed in the opening **O2**. The second conductive plug **188'** is extending from the back side **104** of the first semiconductor substrate **100** into the first semiconductor substrate **100**, and electrically connected with the first conductive plug **131C'**. The second barrier layer **186'** is laterally disposed between the second conductive plug **188'** and the blanket dielectric layer **161**. The second barrier layer **186'** is penetrating through the first dielectric layer **131A'** and in contact with the first barrier layer **131B'**. The blanket dielectric layer **161** is laterally disposed between the second conductive plug **188'** and the first semiconductor substrate **100** and between the second barrier layer **186'** and the first semiconductor substrate **100**.

[0129] In this embodiment, a TSV **12a** includes the first dielectric layer **131A'**, the first barrier layer **131B'**, the first conductive plug **131C'**, the second conductive plug **188'**, the second barrier layer **186'**, and the blanket dielectric layer **161**. In some embodiments, a depth **Y1** of the TSV **12a** beyond the front side **102** of the first semiconductor substrate **100** is less than or equal to a thickness **Z1** of the first ILD layer **132**. In some instances, the depth **Y1** is approximately equal to the thickness of the first ILD layer **132**.

[0130] The conductive via **137** is located above and electrically connected with the TSV **12a**. In this embodiment, since the TSV **12a** does not extend through the ILD structure **130**, the resulting TSV **12a** may have a smaller depth.

[0131] Then, a backside circuit is formed on the back side **104** of the first semiconductor substrate **100**, as shown in FIG. 5M. The process for forming a backside circuit can be referred to in FIGS. 2G to 2I and their related descriptions.

[0132] Finally, the first semiconductor substrate **10C** shown in FIG. 5M is bonded with the second semiconductor **20** as shown in FIG. 5N. The bonding process can be referred to in FIG. 2J and the related descriptions.

[0133] The image sensors **IS** are electrically connected to the transistors **T1**, **T2** via the interconnect structures **140**, **240**. In some embodiments, reducing the distance between the TSV **12a** and the transistor **T1** may reduce the signal transmission path between the image sensor **IS** and the corresponding TSV **12a**, thereby improving the RC delay problem. In some embodiments, the TSV **12a** is overlapping with the pixel area composed of an array of the image sensors **IS**. For example, the TSV **12a** is overlapping with the second semiconductor substrate **200** between two adjacent image sensors **IS** or overlapping with the image sensor **IS**.

[0134] In some embodiments, the image sensor device **4** includes the second semiconductor structure **20** and the first semiconductor structure **10C**, but this disclosure is not limited thereto. In other embodiments, the image sensor device **4** further includes other semiconductor structures, such as ASIC, bonded to the back side of the first semiconductor structure **10C**. In some embodiments, optical elements such as color filters, lens, or other optical elements may be placed on the back side of the second semiconductor structure **20** facing away from the first

semiconductor structure **10C**, but this disclosure is not limited thereto.

[0135] FIG. **6** is a schematic cross-sectional view of an embodiment first semiconductor structure **10D** in accordance with various embodiments. The first semiconductor structure **10D** in FIG. **6** is similar to the first semiconductor structure **10C** in FIG. **5M**. The differences between the two are explained below.

[0136] In the first semiconductor structure **10C** of FIG. **5M**, the etching process for forming the opening **O2** in the first semiconductor substrate **100** stops on the first barrier layer **131B'**, so that the subsequently formed blanket dielectric layer **161** contacts the bottom surface of the first barrier layer **131B'** (as shown in FIG. **5J**). However, in the first semiconductor structure **10D** of FIG. **6**, the etching process for forming the opening **O2** stops on the first dielectric layer **131A'**, so that the blanket dielectric layer (referred to as the second dielectric layer **161'** after the CMP process) is separated from the first barrier layer **131B'** by the first dielectric layer **131A'**. Therefore, in the process of the first semiconductor structure **10D**, after the blanket dielectric layer is formed, it is necessary to etch both the blanket dielectric layer **161** and the first dielectric layer **131A'** to expose the first barrier layer **131B'**.

[0137] In addition, in the first semiconductor structure **10D** of FIG. **6**, the CMP process for grinding the metal material in the opening **O2** (similar to the step of FIG. **5L**) stops on the buffer layer **150**, and the materials beyond the buffer layer **150** are removed in the CMP process. In the first semiconductor structure **10D** of FIG. **6**, a portion of the blanket dielectric layer beyond the buffer layer **150** is removed during the CMP process, remaining the second dielectric layer **161'**. In some embodiments, a portion of the mask layer beyond the buffer layer **150** is removed during the CMP process, remaining the mask layer **171'**. In some embodiments, the mask layer **171'** is laterally disposed between the second barrier layer **186'** and the second dielectric layer **161'**.

[0138] In some embodiments, the first semiconductor structure **10D** and the second semiconductor structure **20** are bonded together using a method similar to that disclosed in FIG. **2J** to obtain an image sensor device.

[0139] Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

[0140] In accordance with an embodiment, an image sensor device includes a first semiconductor structure and a second semiconductor structure bonding with the first semiconductor structure. The first semiconductor structure includes a first semiconductor substrate and logic devices. The logic devices are disposed on the first semiconductor substrate. A through-substrate via is penetrating through the first semiconductor substrate and located between two of the logic devices. A second semiconductor structure includes a second semiconductor substrate and image sensors. The image sensors are disposed in a pixel area of the second semiconductor substrate. At least one of the image sensors is electrically to at least one of the two of the logic devices.

[0141] In accordance with another embodiment, a semiconductor structure includes a semiconductor substrate a transistor, an interlayer dielectric structure, a source/drain contact via, a conductive via, and a through-substrate via. The transistor is disposed on the semiconductor substrate. The interlayer dielectric structure is disposed over the transistor. A source/drain contact via is embedded in the interlayer dielectric structure and extending from a top surface of the interlayer dielectric structure to a source/drain feature of the transistor. A conductive via is embedded in the interlayer dielectric structure and extending from the top surface of the interlayer dielectric structure into the interlayer dielectric structure. A height of the source/drain contact via is

greater than a height of the conductive via. The through-substrate via is penetrating through the semiconductor substrate and electrically connected with the conductive via.

[0142] In accordance with still another embodiment, an image sensor device includes a first semiconductor substrate, a transistor, a first interlayer dielectric layer, a second interlayer dielectric layer, a source/drain contact via, and a through-substrate via. The first semiconductor substrate includes a front side and a back side opposite to the front side. The transistor is disposed on the front side of the first semiconductor substrate. The first interlayer dielectric layer is disposed over the front side of the first semiconductor substrate. The second interlayer dielectric layer is disposed over the first interlayer dielectric layer. The source/drain contact via is extending through the first interlayer dielectric layer and the second interlayer dielectric layer. The source/drain contact via is electrically connected with the transistor. The through-substrate via is extending from the back side of the first semiconductor substrate to beyond the front side of the first semiconductor substrate. A depth of the through-substrate via beyond the front side of the first semiconductor substrate is less than or equal to a thickness of the first interlayer dielectric layer.

[0143] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An image sensor device, comprising: a first semiconductor structure, comprising: a first semiconductor substrate; logic devices, disposed on the first semiconductor substrate; and a through-substrate via, penetrating through the first semiconductor substrate and located between two of the logic devices; and a second semiconductor structure bonding with the first semiconductor structure, wherein the second semiconductor structure comprising: a second semiconductor substrate; and image sensors, disposed in a pixel area of the second semiconductor substrate, wherein at least one of the image sensors is electrically to at least one of the two of the logic devices.
2. The image sensor device of claim 1, wherein the through-substrate via is overlapping with the pixel area.
3. The image sensor device of claim 1, wherein the first semiconductor structure further comprises: a gate electrode layer, comprises a gate electrode of the at least one of the two of the logic devices and a conductive pad; a gate insulation layer, located between the gate electrode and the first semiconductor substrate and between the conductive pad and the first semiconductor substrate, wherein the through-substrate via is penetrating through the gate insulation layer and in contact with the conductive pad.
4. The image sensor device of claim 1, wherein the first semiconductor structure further comprises: an interlayer dielectric structure, disposed over the logic devices; a source/drain contact via, embedded in the interlayer dielectric structure and electrically connected with a source/drain feature of the at least one of the two of the logic devices; a conductive pad, embedded in the interlayer dielectric structure, wherein the through-substrate via is extending into the interlayer dielectric structure and in contact with the conductive pad; and a conductive via, embedded in the interlayer dielectric structure and disposed over the conductive pad, wherein a height of the source/drain contact via is greater than a height of the conductive via.
5. The image sensor device of claim 4, wherein the first semiconductor structure further comprises:

an interconnect structure, disposed over the interlayer dielectric structure and electrically connected with the conductive via and the source/drain contact via.

6. The image sensor device of claim 4, wherein a portion of the interlayer dielectric structure is disposed between the conductive pad and the first semiconductor substrate.

7. The image sensor device of claim 1, wherein the through-substrate via comprises: a conductive plug, embedded in the first semiconductor substrate; a barrier layer, surrounding the conductive plug; a dielectric layer, disposed between the barrier layer and the first semiconductor substrate; and a mask layer, disposed between the dielectric layer and the barrier layer, wherein a portion of the barrier layer is in contact with the dielectric layer, and another portion of the barrier layer is in contact with the mask layer.

8. The image sensor device of claim 1, wherein the through-substrate via comprises: a first conductive plug, extending from a front side of the first semiconductor substrate into the first semiconductor substrate, wherein the logic devices are disposed on the front side of the first semiconductor substrate; a second conductive plug, extending from a back side of the first semiconductor substrate into the first semiconductor substrate, and electrically connected with the first conductive plug; a first dielectric layer, laterally disposed between the first conductive plug and the first semiconductor substrate; and a second dielectric layer, laterally disposed between the second conductive plug and the first semiconductor substrate.

9. The image sensor device of claim 8, wherein the through-substrate via comprises: a first barrier layer, laterally disposed between the first conductive plug and the first dielectric layer, wherein the second dielectric layer is in contact with the first barrier layer; and a second barrier layer, laterally disposed between the second conductive plug and the second dielectric layer, wherein the second barrier layer is in contact with the first barrier layer.

10. A semiconductor structure, comprising: a semiconductor substrate; a transistor, disposed on the semiconductor substrate; an interlayer dielectric structure, disposed over the transistor; a source/drain contact via, embedded in the interlayer dielectric structure and extending from a top surface of the interlayer dielectric structure to a source/drain feature of the transistor; a conductive via, embedded in the interlayer dielectric structure and extending from the top surface of the interlayer dielectric structure into the interlayer dielectric structure, wherein a height of the source/drain contact via is greater than a height of the conductive via; and a through-substrate via, penetrating through the semiconductor substrate and electrically connected with the conductive via.

11. The semiconductor structure of claim 10, further comprises: a gate electrode layer comprising a conductive pad and a gate electrode of the transistor, wherein the conductive pad is located between the through-substrate via and the conductive via; and a gate insulation layer, located between the gate electrode and the semiconductor substrate and between the conductive pad and the semiconductor substrate, wherein the through-substrate via is penetrating through the gate insulation layer.

12. The semiconductor structure of claim 11, wherein a material of the conductive pad comprises polysilicon.

13. The semiconductor structure of claim 10, wherein the interlayer dielectric structure comprises: a first interlayer dielectric layer, disposed over the transistor; and a second interlayer dielectric layer, disposed over the first interlayer dielectric layer; and a conductive pad, located between the first interlayer dielectric layer and the second interlayer dielectric layer and between the through-substrate via and the conductive via.

14. The semiconductor structure of claim 13, wherein the source/drain contact via is penetrating through the first interlayer dielectric layer and the second interlayer dielectric layer, and the conductive via is penetrating through the second interlayer dielectric layer.

15. The semiconductor structure of claim 10, wherein the through-substrate via comprises: a first conductive plug, extending from the interlayer dielectric structure into the semiconductor substrate, wherein the conductive via is landed on the first conductive plug; and a second conductive plug,

extending from a back side of the semiconductor substrate into the semiconductor substrate, and electrically connected with the first conductive plug; a first dielectric layer, laterally disposed between the first conductive plug and the semiconductor substrate; and a second dielectric layer, laterally disposed between the second conductive plug and the semiconductor substrate, wherein the second dielectric layer is penetrating through the first dielectric layer.

16. An image sensor device, comprising: a first semiconductor substrate comprising a front side and a back side opposite to the front side; a transistor, disposed on the front side of the first semiconductor substrate; a first interlayer dielectric layer, disposed over the front side of the first semiconductor substrate; a second interlayer dielectric layer, disposed over the first interlayer dielectric layer; and a source/drain contact via, extending through the first interlayer dielectric layer and the second interlayer dielectric layer, wherein the source/drain contact via is electrically connected with the transistor; a through-substrate via, extending from the back side of the first semiconductor substrate to beyond the front side of the first semiconductor substrate, wherein a depth of the through-substrate via beyond the front side of the first semiconductor substrate is less than or equal to a thickness of the first interlayer dielectric layer.

17. The image sensor device of claim 16, further comprises: a second semiconductor substrate, overlapping with the first semiconductor substrate; and a photodiode, disposed in a pixel area of the second semiconductor substrate and electrically connected with the transistor, wherein the through-substrate via is overlapping with the pixel area.

18. The image sensor device of claim 16, further comprises: an interlayer conductive layer, located between the first interlayer dielectric layer and the second interlayer dielectric layer, wherein the through-substrate via is extending into the first interlayer dielectric layer and in contact with the interlayer conductive layer.

19. The image sensor device of claim 16, further comprises: a conductive pad, located between the first interlayer dielectric layer and the first semiconductor substrate; an insulation structure, located between the conductive pad and the first semiconductor substrate, wherein a sidewall of the insulation structure is aligned with a sidewall of the conductive pad, and the through-substrate via is penetrating through the insulation structure and in contact with the conductive pad.

20. The image sensor device of claim 16, wherein the through-substrate via comprises: a first conductive plug, extending from a top surface of the first interlayer dielectric layer into the first semiconductor substrate; and a second conductive plug, extending from the back side of the first semiconductor substrate into the first semiconductor substrate, and electrically connected with the first conductive plug.
